Philip-Chandan Imaging Pipeline

TCGA-BRCA longitudinal MRI: discovery through raw export

QBIHack breast-cancer-sim | Generated 2026-06-27

Philip-Chandan owns the radiomics pipeline: pull real breast MRI from TCIA, validate and stack DICOM into 3D numpy arrays, and hand raw volumes to Vinesh for PDE simulation. Resample, normalize, and segmentation are Vinesh's scope (handoff_contract.json v1.0.0, Option B).

1. Patient selection criteria

We needed real TCGA-BRCA cases for a Luminal A vs Basal-like demo with longitudinal MRI, not a single timepoint. Each candidate had to pass several gates:

Criterion	Why it matters
MRI on TCIA	Many TCGA patients have genomics but no public imaging
Longitudinal MR (>=2 study dates)	Baseline + follow-up for growth comparison
PAM50 subtype match	Luminal A and Basal-like labels align with cBioPortal
Genomics for Praneeth	ER/PR status and survival on GDC/cBioPortal
Usable contrast series	Post-contrast T1 (e.g. VIBRANT) for tumor visibility

Our first picks failed: TCGA-BH-A0BR and TCGA-A2-A04P had no MRI on TCIA. Only 19 of 139 TCGA-BRCA patients with MRI on TCIA have multiple MR studies. Patient selection was a real constraint.

Rev2 primary pair (cohort/cohort.json)

Subtype	TCGA ID	Baseline	Follow-up
Luminal A	TCGA-AR-A1AX	2002-09-12	2003-09-24 (~12 mo)
Basal-like	TCGA-AR-A1AQ	2001-11-21	2003-05-07 (~17 mo)

2. Finding patients - cohort_discovery.py

Before locking IDs in cohort.json, we built cohort/cohort_discovery.py to automate discovery and validation against public APIs:

- TCIA NBIA REST API - list MR series, group by study date, flag longitudinal cases, prefer contrast series
- cBioPortal - PAM50 subtype, ER/PR, overall survival (aligned with Praneeth's genomics fields)

CLI workflows

```
python .../cohort/cohort_discovery.py audit
python .../cohort/cohort_discovery.py find-longitudinal --subtype "Luminal A"
python .../cohort/cohort_discovery.py recommend-pair
```

Each patient gets a structured report: imaging availability, study span, PAM50 match, missing genomics fields, and a pass/fail ok flag. This tool justified the rev2 pivot and gave Praneeth the same TCGA barcodes for GDC queries.

3. Download - download_tcia.py

Once patients are locked, download_tcia.py pulls DICOM into a consistent layout:

```
data/raw/tcia/
  luminal_a/TCGA-AR-A1AX/
    2002-09-12/   # baseline
    2003-09-24/   # follow-up
  basal/TCGA-AR-A1AQ/
    2001-11-21/
```

```
    2003-05-07/
```

For each study date the downloader:

- Lists MR series in collection TCGA-BRCA
- Picks the best series - prefers post-contrast (+C, VIBRANT, T1); deprioritizes calibration/localizer scans
- Downloads via idc-index (primary) with tcia-utils NBIA fallback

```
python .../download_tcia.py --all-primary --longitudinal
```

4. Validate - validate_series() in tcia_extractor.py

Download alone is not enough. Before export, validate_series(dicom_dir) checks the local series:

Check	What we catch
DICOM slices present	Empty or non-image folders
Consistent Rows/Columns	Mixed slice dimensions
Unique InstanceNumber	Duplicate or corrupted ordering
SimpleITK read succeeds	Metadata OK but ITK cannot stack
Slice count, shape, spacing_mm	Recorded in validation report for QC and sidecar JSON

extract_volume() refuses to run if validation fails. Export scripts call validate_series first and abort with explicit errors.

Visual QC

qc_slice_plot.py writes a middle-slice PNG per volume to data/qc/slice-plots-philip-chandan/ so a human can confirm anatomy looks real before handoff.

5. Export - raw volumes for Vinesh

After validation passes, we export raw 3D arrays, not PDE-ready volumes. Resample, normalize, and segmentation are Vinesh's scope per handoff_contract.json.

- Single case: export_raw_extract.py
- Batch (all primaries x timepoints): export_all_raw.py

For each patient x timepoint slug (e.g. luminal_a_TCGA-AR-A1AX_baseline):

- Resolve DICOM path from cohort.json
- Run validate_series
- extract_volume_with_spacing() - SimpleITK DICOM to float32 numpy, shape (Z, Y, X), native spacing preserved
- Write .npy, JSON sidecar, and QC plot

Output	Path
Raw volume	data/processed/raw-extract-philip-chandan/{slug}.npy
Metadata sidecar	.../{slug}.json (shape, spacing_mm, contract_version)
QC plot	data/qc/slice-plots-philip-chandan/{slug}_mid-z.png
Manifest	manifest.json v1.1.0 - subtype, timepoint, slug, paths

Handoff contract - what we export vs what we do not

Philip-Chandan (raw extract)	Vinesh (PDE input)
Raw MR intensities, not normalized	Resample/crop to max 64^3, 1 mm spacing
Spacing in JSON sidecar	Normalize to [0, 1]

Axis order (Z, Y, X)	Otsu tumor segmentation

6. End-to-end summary

We defined strict patient criteria (longitudinal MRI on TCIA, PAM50-aligned subtypes, genomics for the team), built cohort_discovery.py to search and audit candidates against TCIA and cBioPortal, locked rev2 primaries after rev1 failed imaging availability, downloaded four DICOM timepoints with contrast-aware series selection, validated each series structurally before extraction and visually via slice QC, then exported four raw .npy volumes plus JSON sidecars and a manifest for downstream PDE simulation and UI integration.

Deliverables

Stage	Deliverable	Count
Discovery	cohort.json rev2 + COHORT.md	2 primaries + backups
Download	DICOM under data/raw/tcia/	4 study folders
Validate	validate_series pass + QC PNGs	4 volumes
Export	Raw .npy + .json + manifest.json	4 slugs

Exported volume shapes (rev2)

Slug	Shape (Z,Y,X)	Spacing (mm)
luminal_a_TCGA-AR-A1AX_baseline	[352, 256, 256]	[3.0, 0.8594, 0.8594]
basal_TCGA-AR-A1AQ_baseline	[464, 256, 256]	[3.0, 0.859375, 0.859375]
luminal_a_TCGA-AR-A1AX_followup	[552, 512, 512]	[2.2, 0.5273, 0.5273]
basal_TCGA-AR-A1AQ_followup	[448, 256, 256]	[3.0, 0.9375, 0.9375]